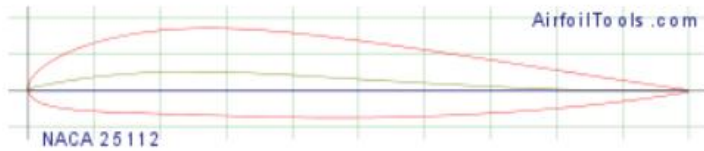


FLOW SIMULATION ON AIRFOIL USING SOLID WORKS

(naca25112-jf) NACA 25112



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Infinity Explorers 2



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TEAM MEMBERS:

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- Mriam Mansour Bishara
- Mariam Emad Kamal
- Yousef Abdullah Abdelhalim

(NACA 25112) Airfoil Report

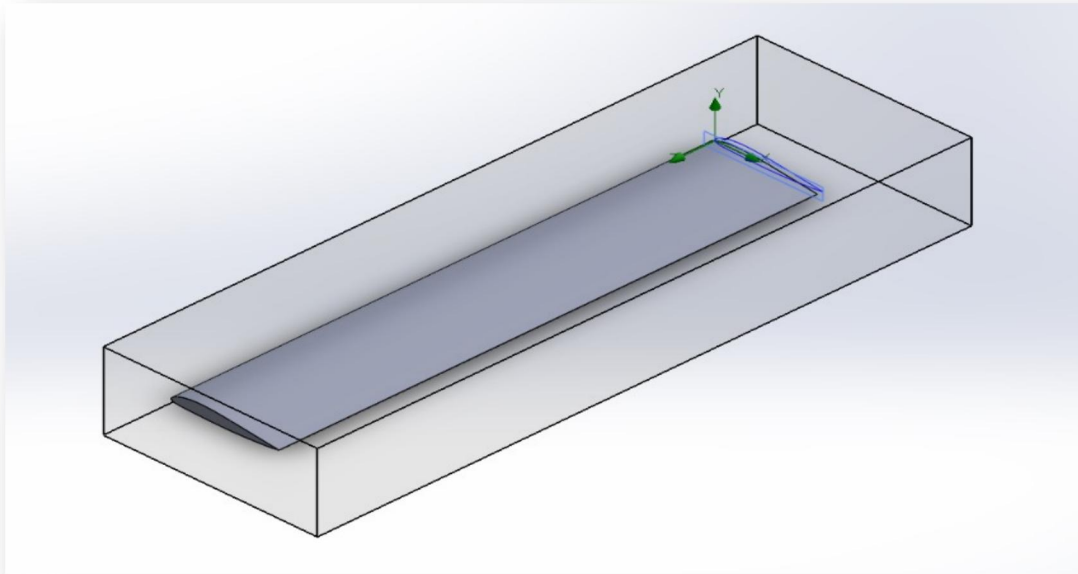


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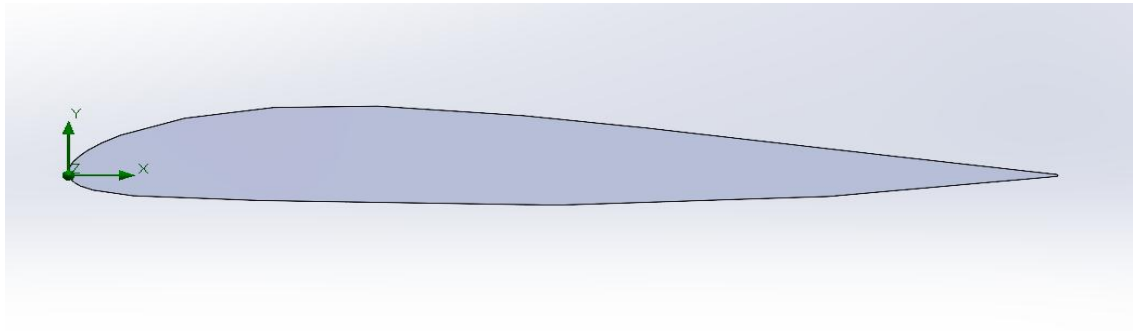
Drawing the airfoil:

Airfoil Data:

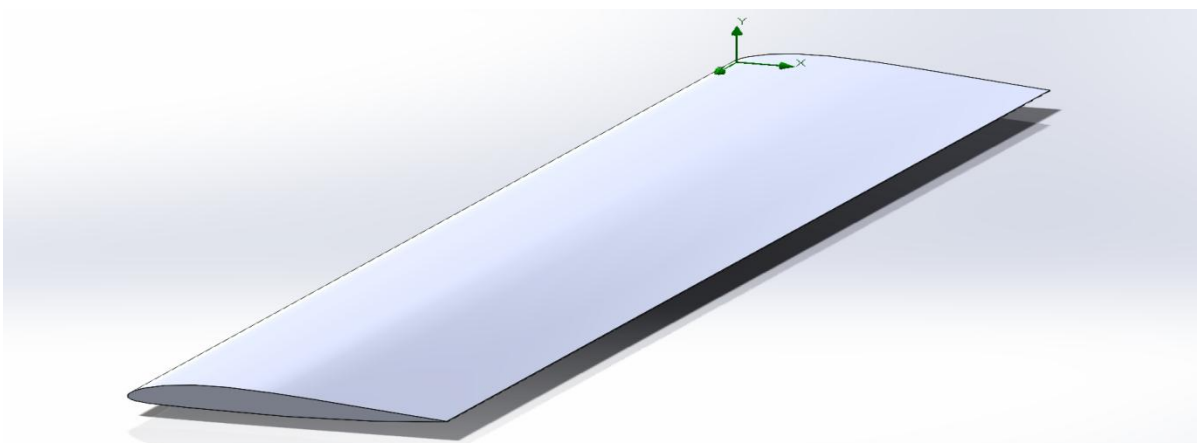
Chord line length= 200 mm.

Length = 1000 mm.

Steps of Drawing:

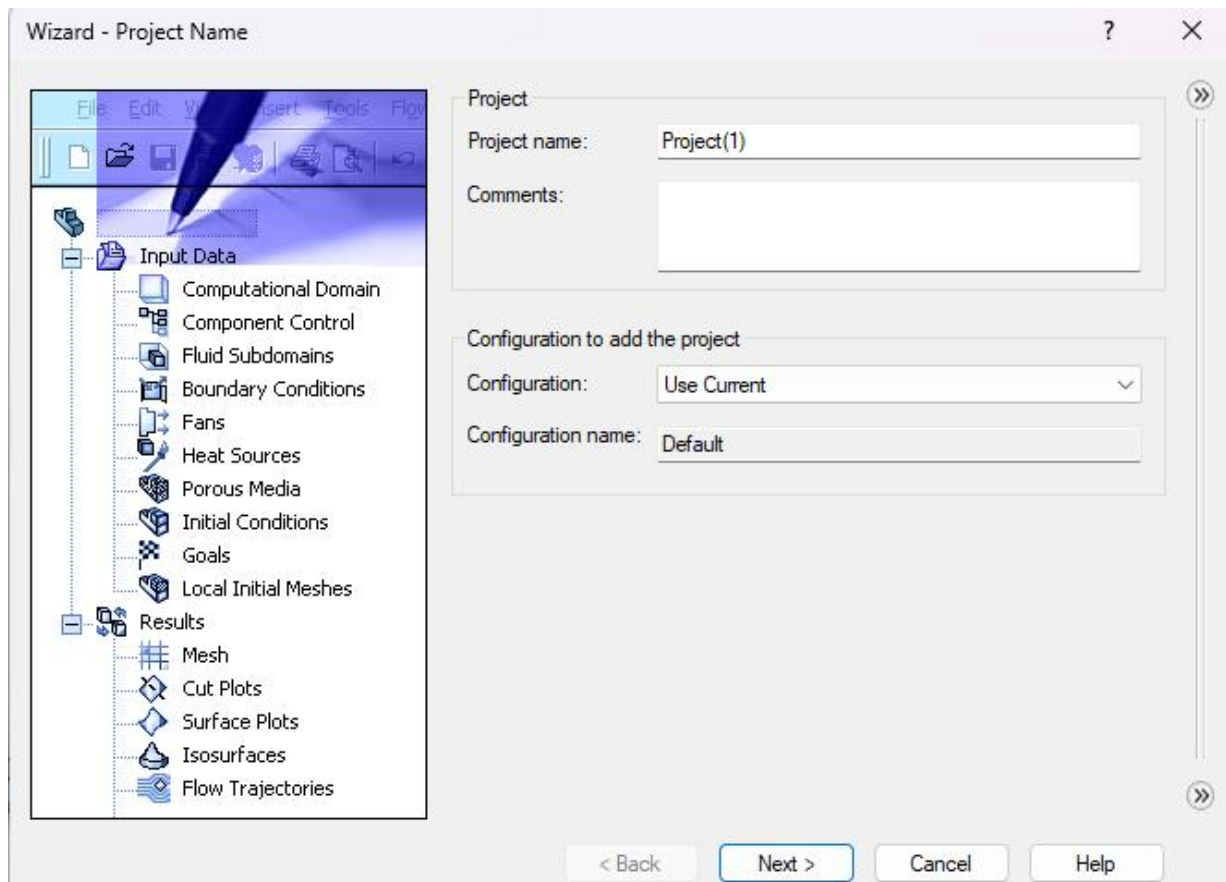


1. Insert the curve via (curve through XYZ points) (NACA 25112) airfoil in our case
2. choose curve from SKETCH after selecting plane
3. Convert entities.
4. Insert new plane from FEATURES, references plane.
5. copy curve to the inserted plane.
6. delete any extra curves.
7. Spam 1 meters by Boss extrude

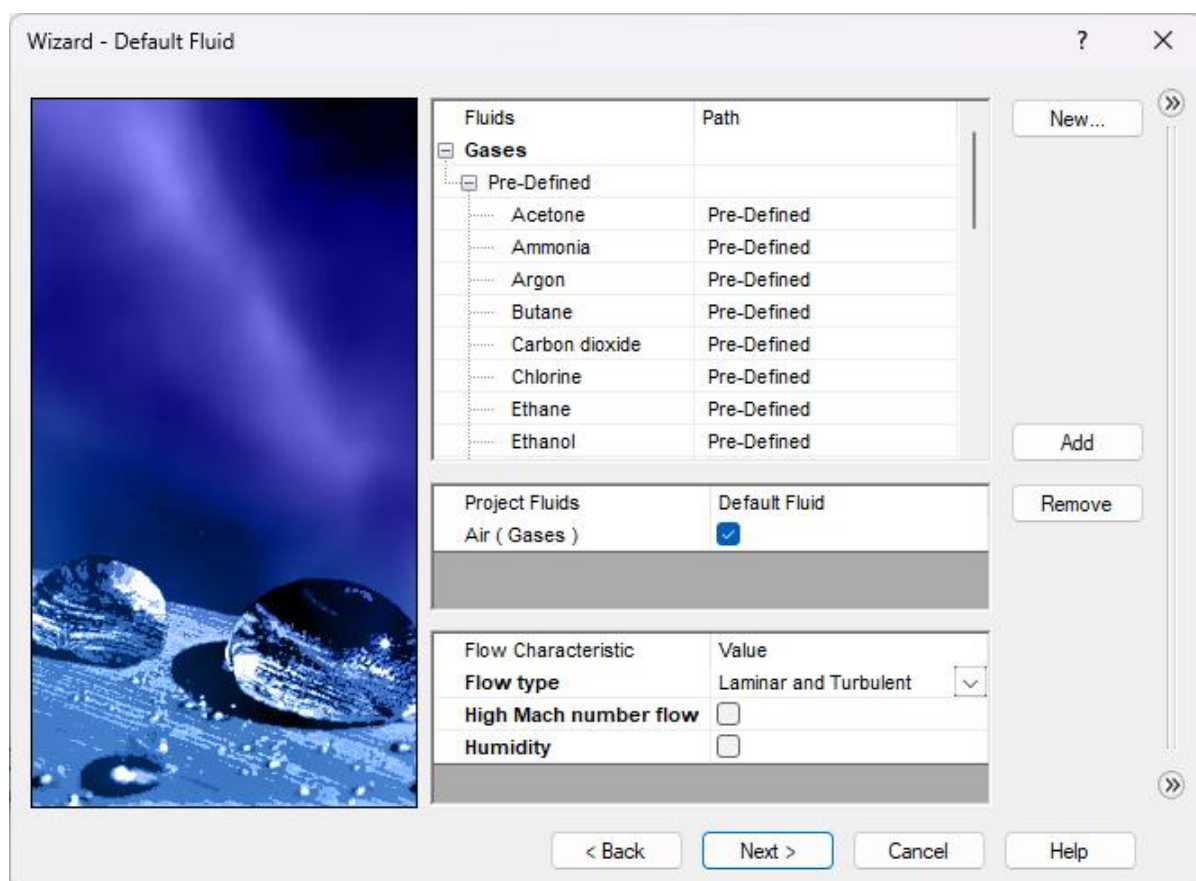


Flow simulation for the airfoil:

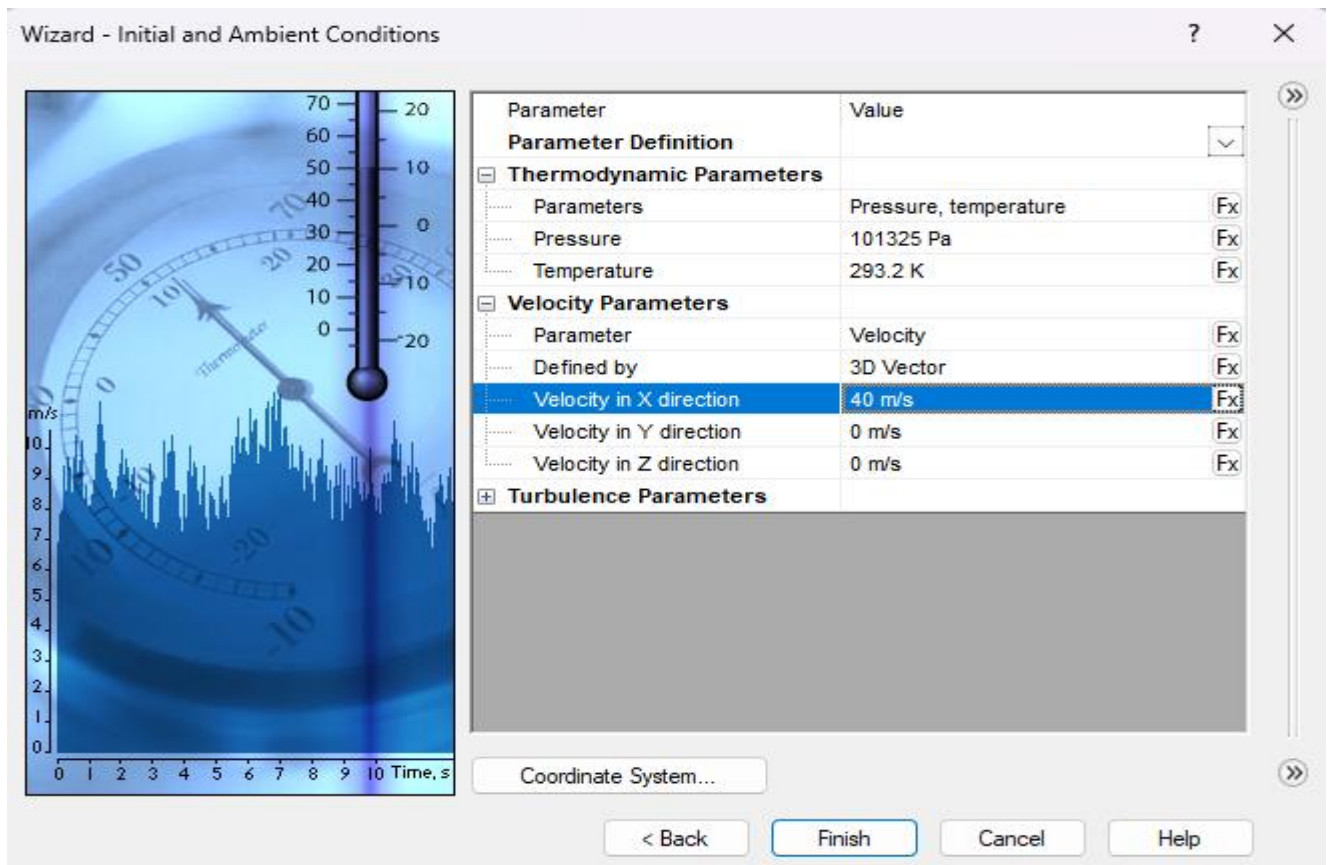
1. from flow simulation choose wizard and name the project.



2. Choose the gas for the simulation (air)



3. Set the velocity in the X direction



4. Click finish

5. Set boundary conditions

- Choose the body of the air foil as face
- Set the roughness to (297.4)

6. Set Goals

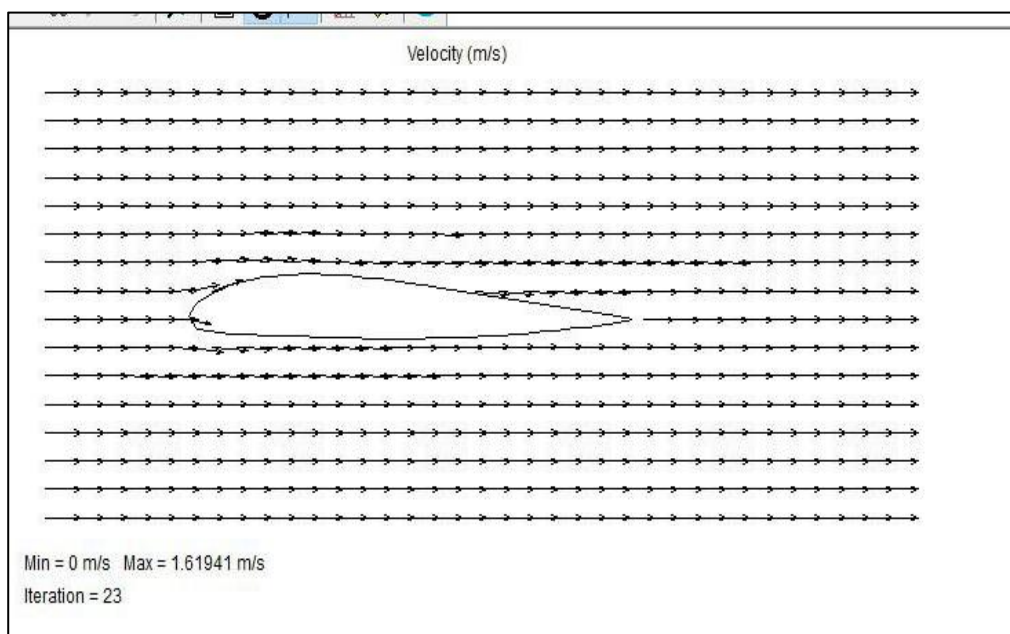
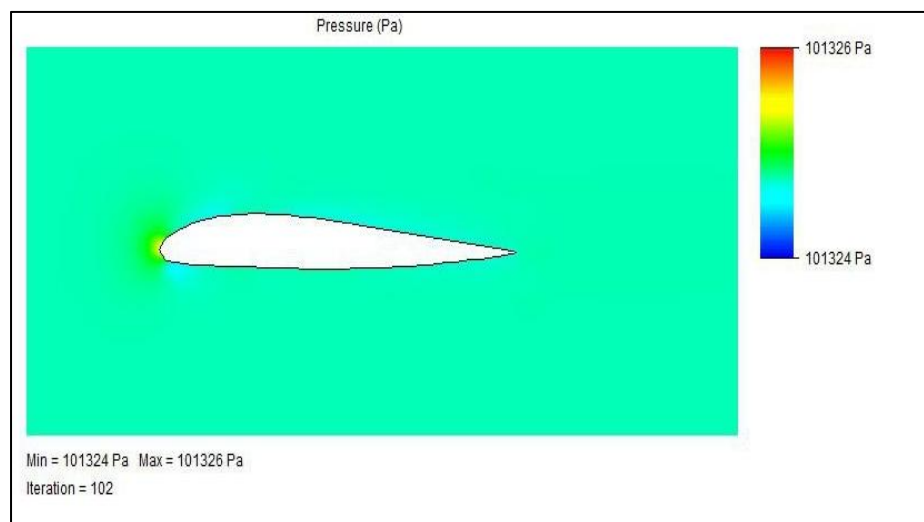
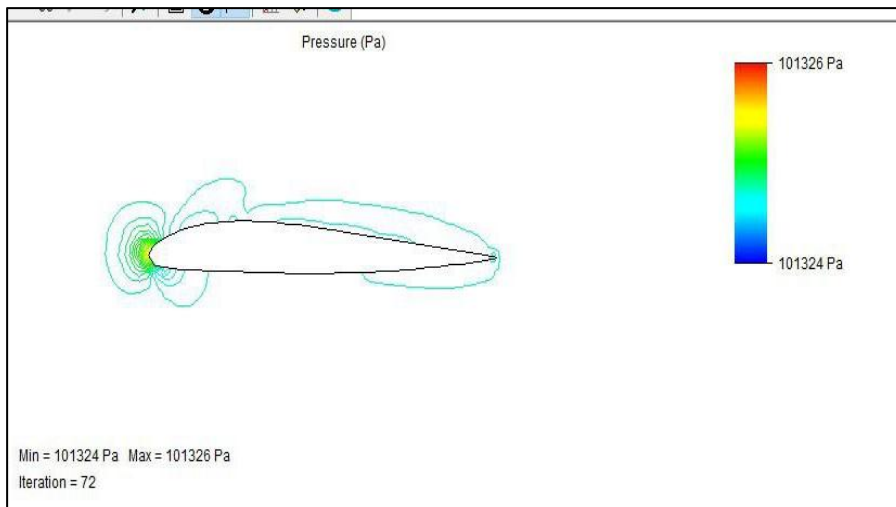
- Surface goals
(static pressure, force X (drag) and Y (lift))
- Equation goals
(set equation for the CL and CD)
- Drag equation
$$2 \cdot \{\text{Drag Force (X)}\} / (1.2 \cdot (V)^2 \cdot 0.2 \cdot 1)$$
- Lift equation
$$2 \cdot \{\text{Left Force (Y)}\} / (1.2 \cdot (V)^2 \cdot 0.2 \cdot 1)$$

7. Set the mesh up to 3 (for simplified analysis)

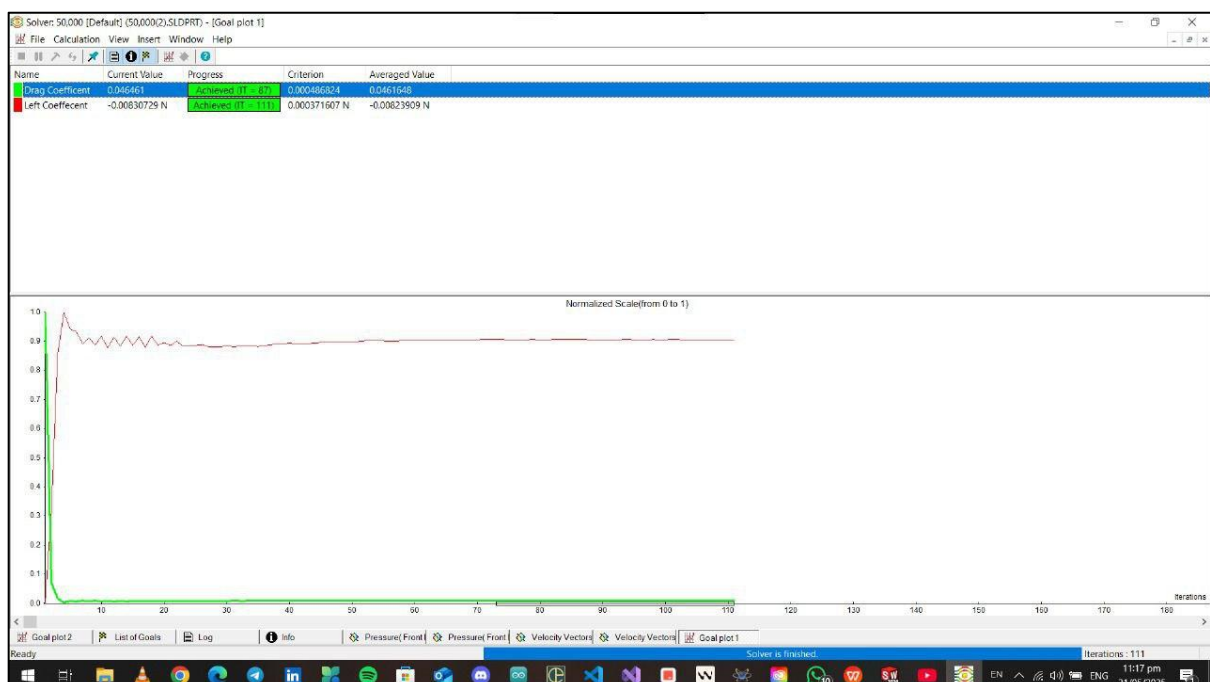
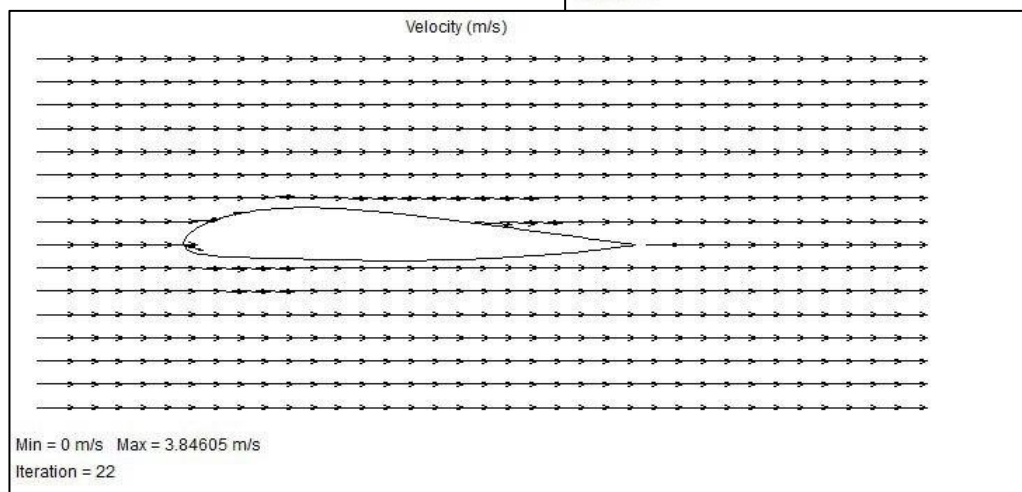
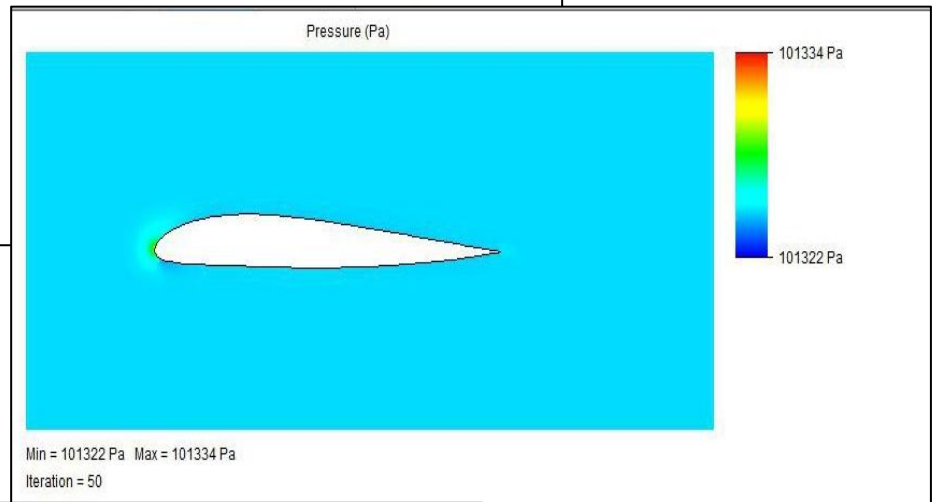
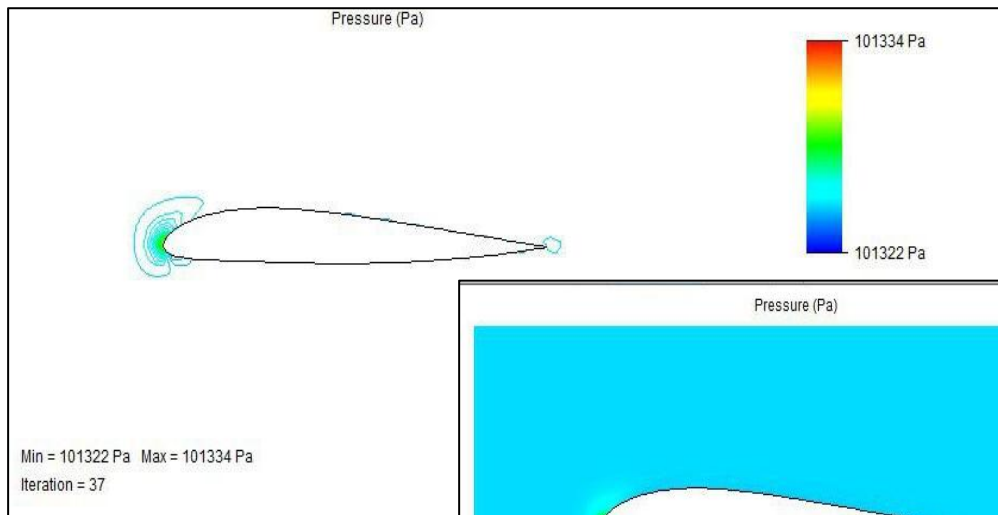
8. Run

Output Results

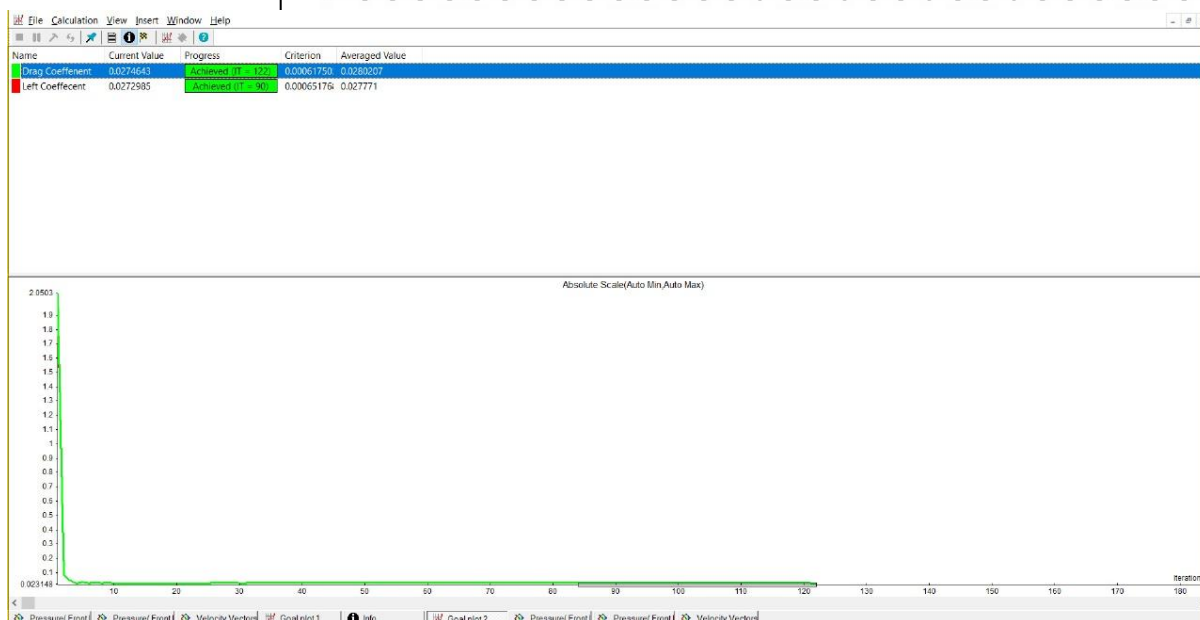
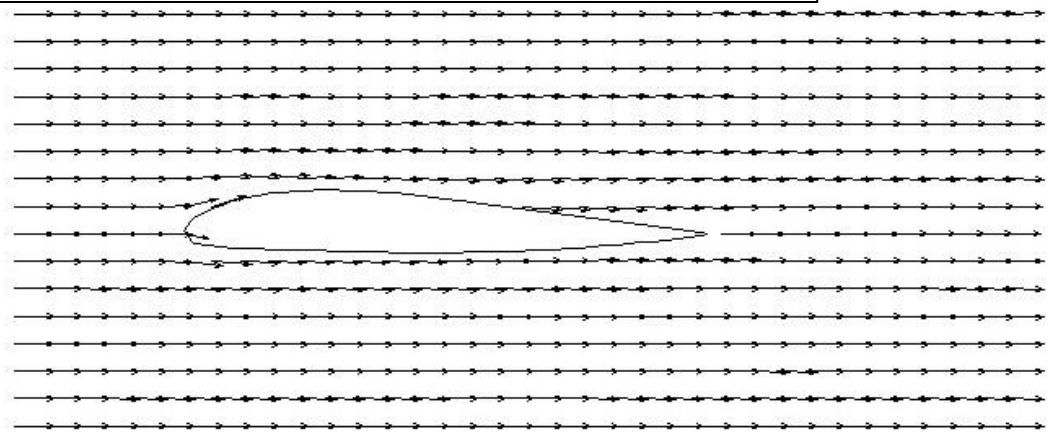
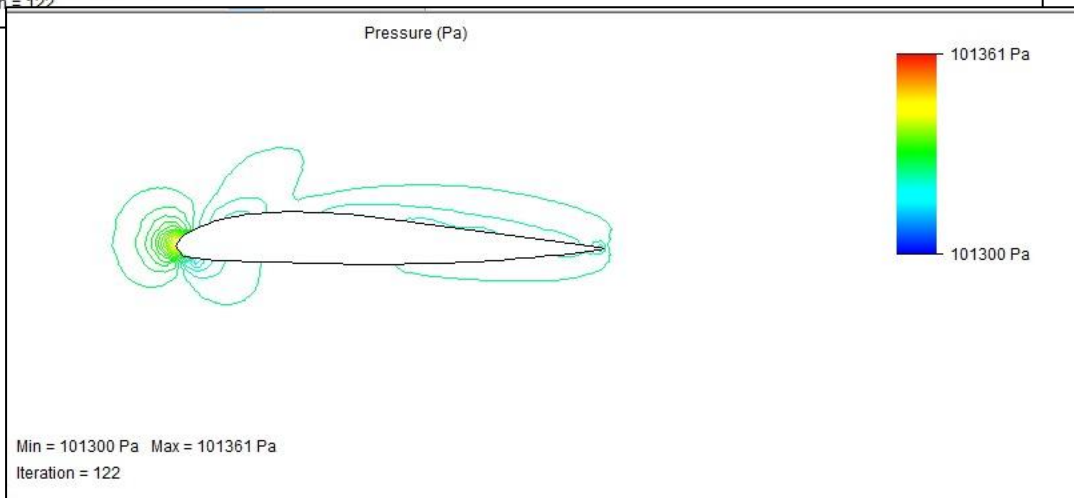
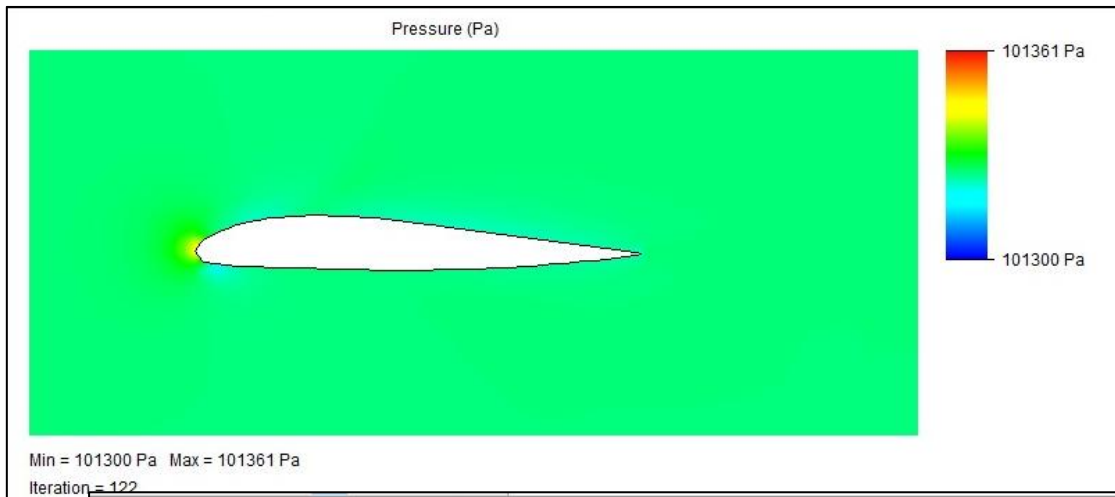
20K Re



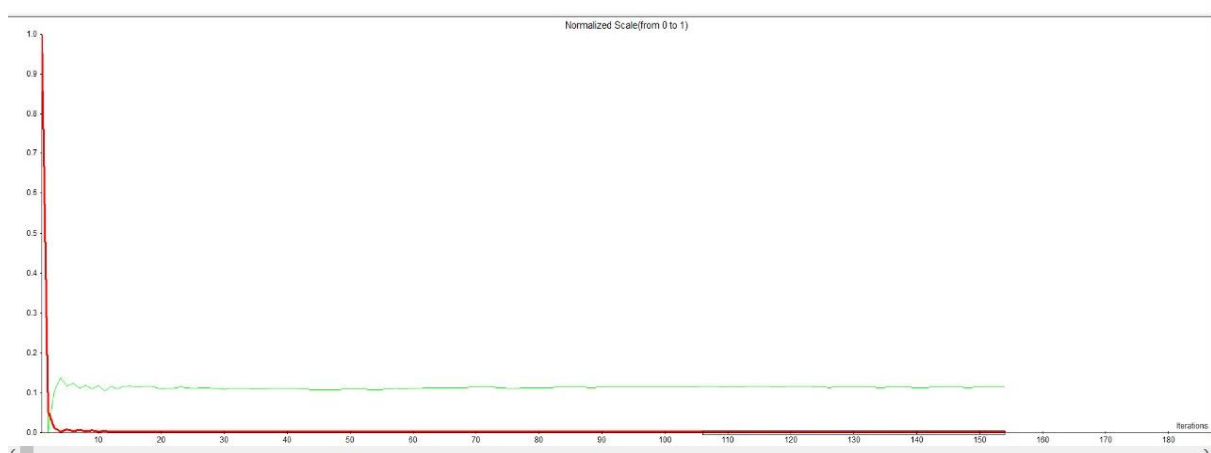
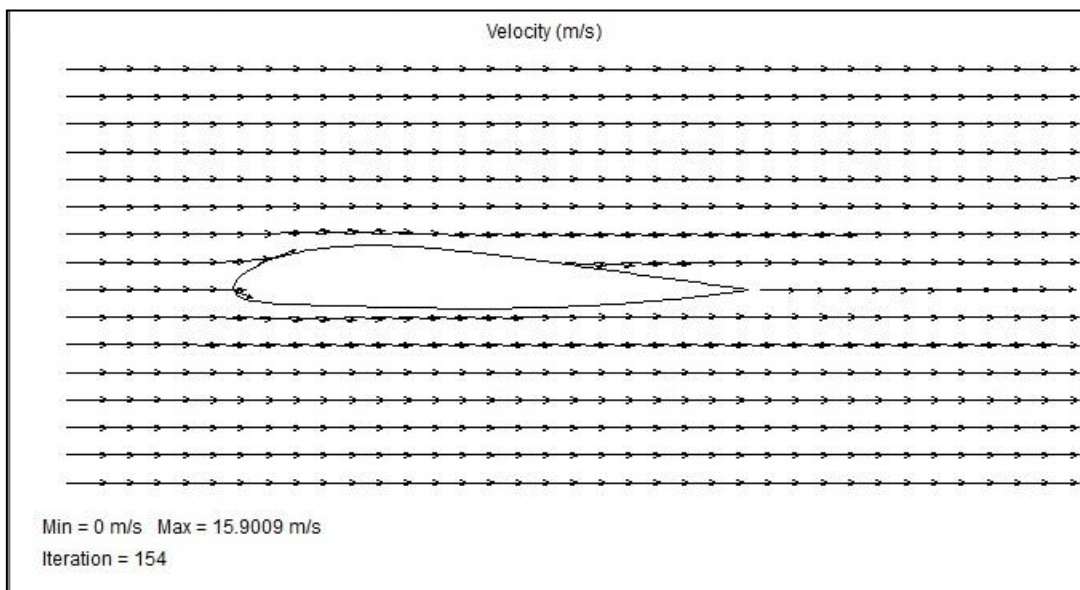
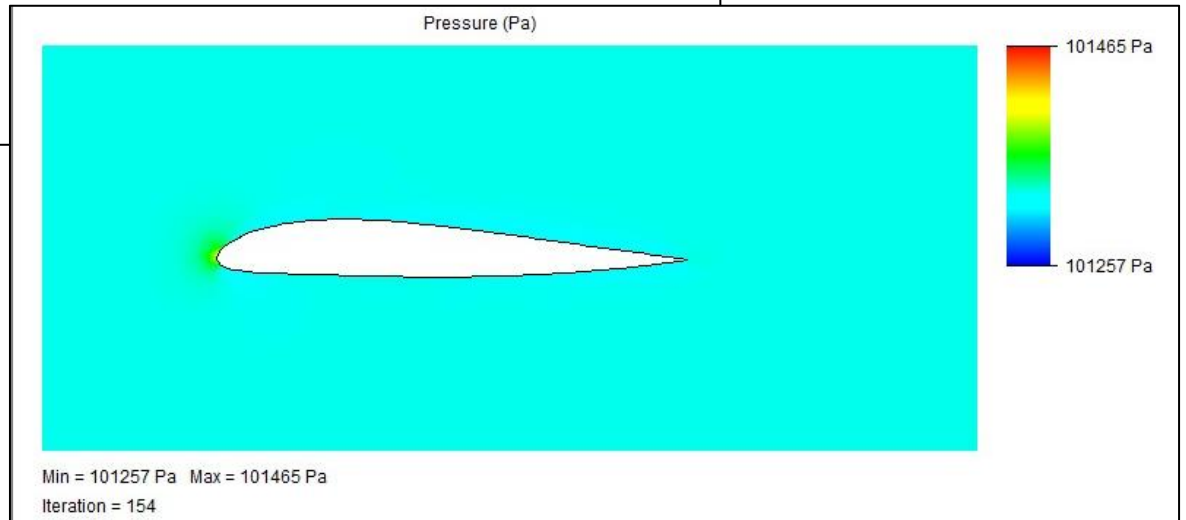
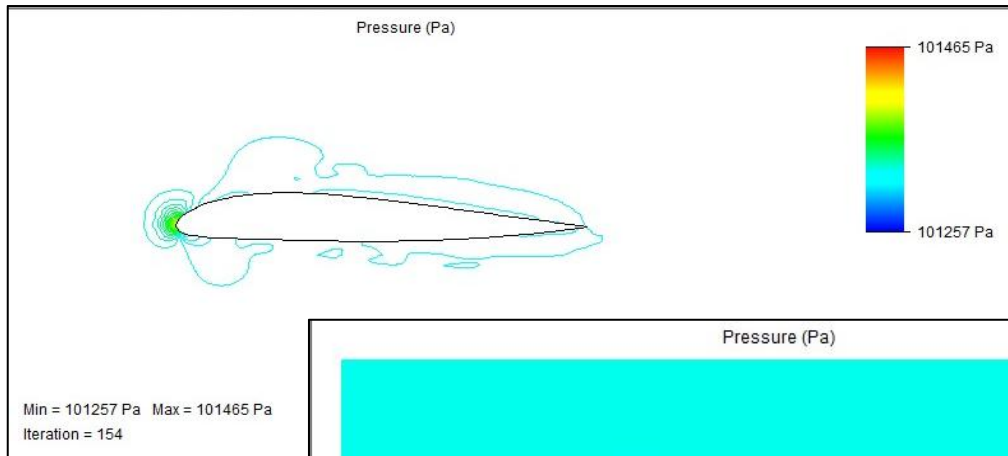
50K Re



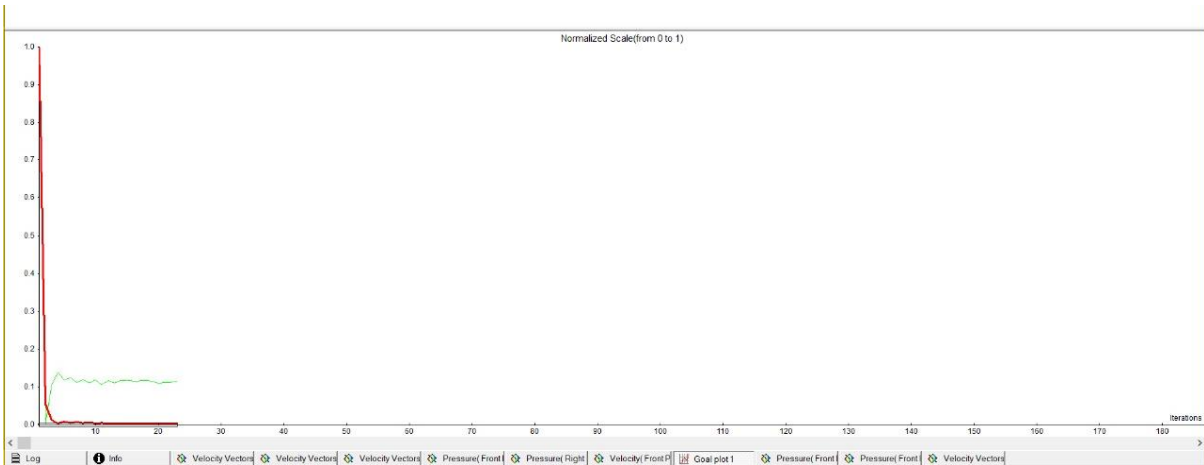
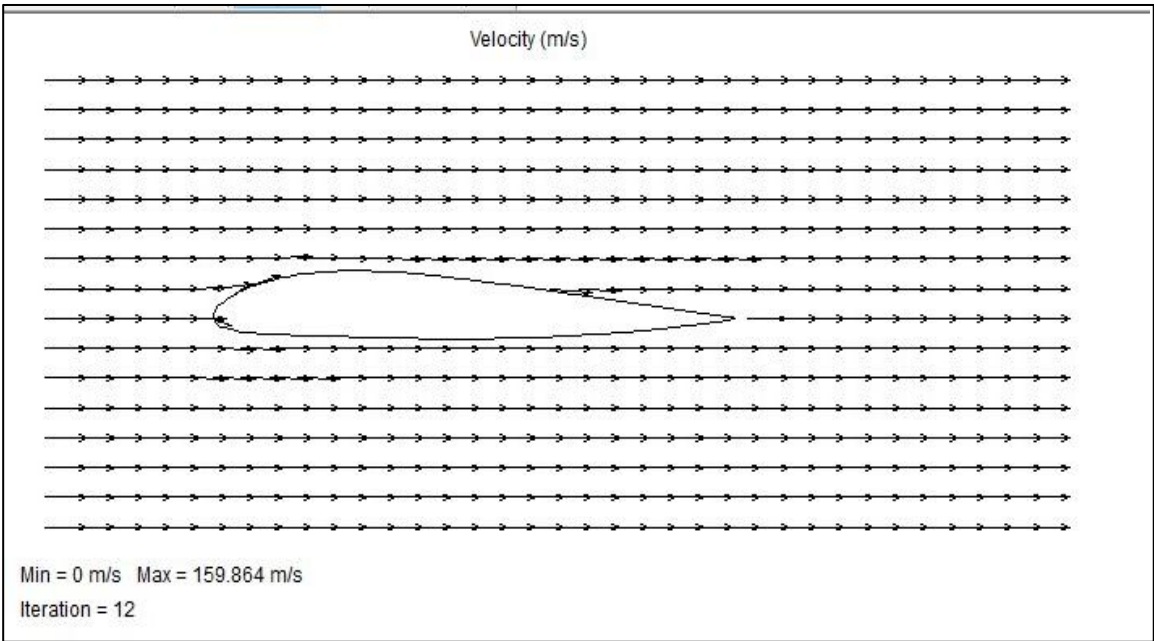
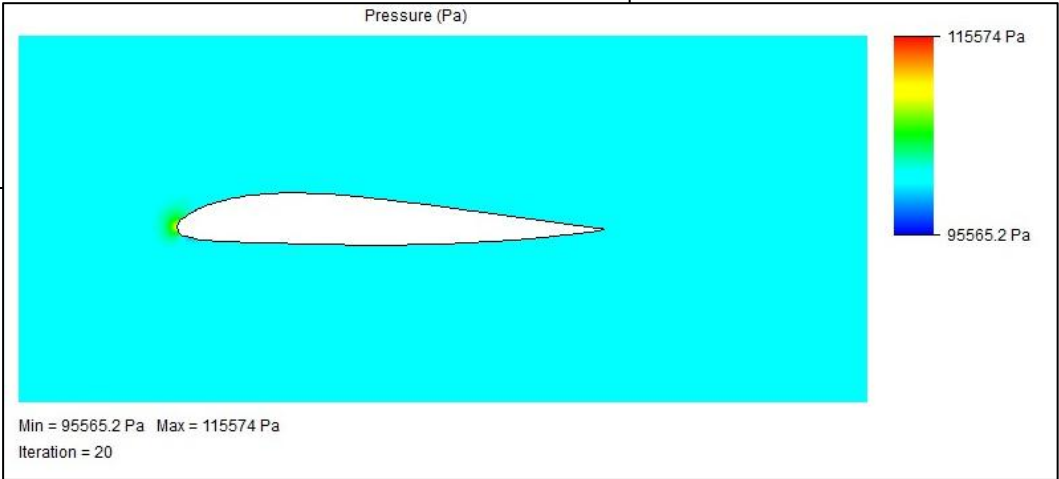
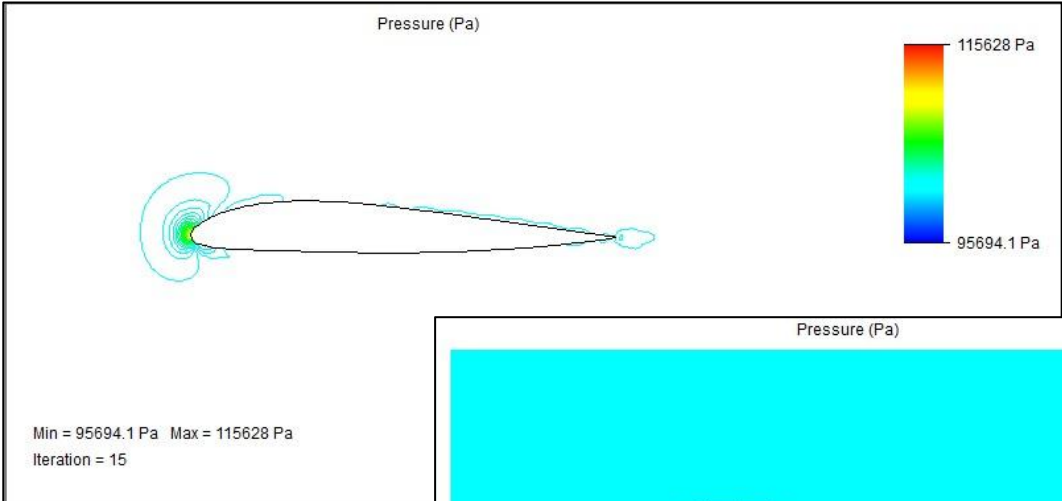
100K
Re



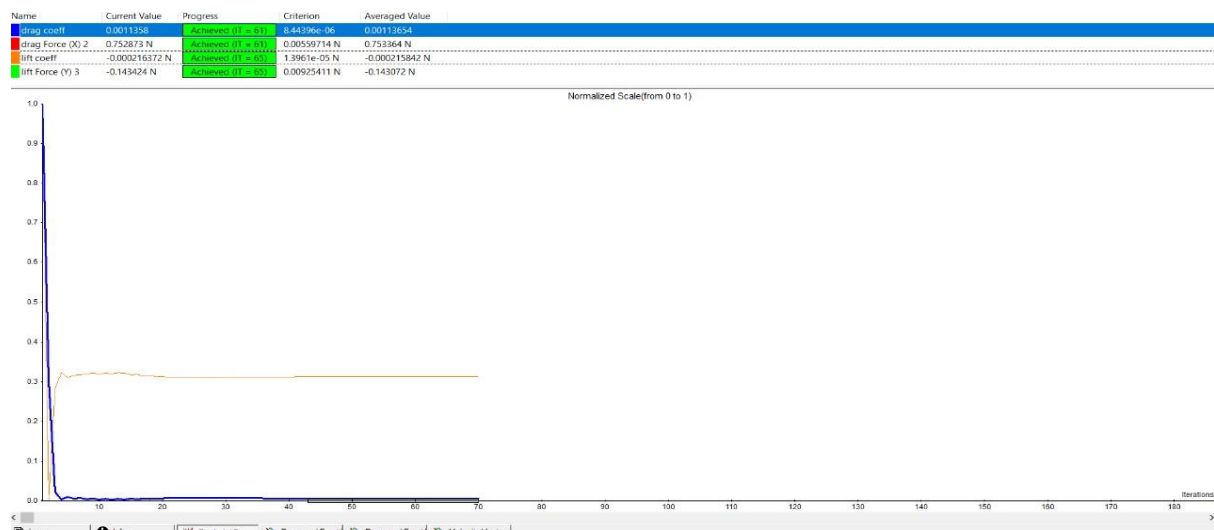
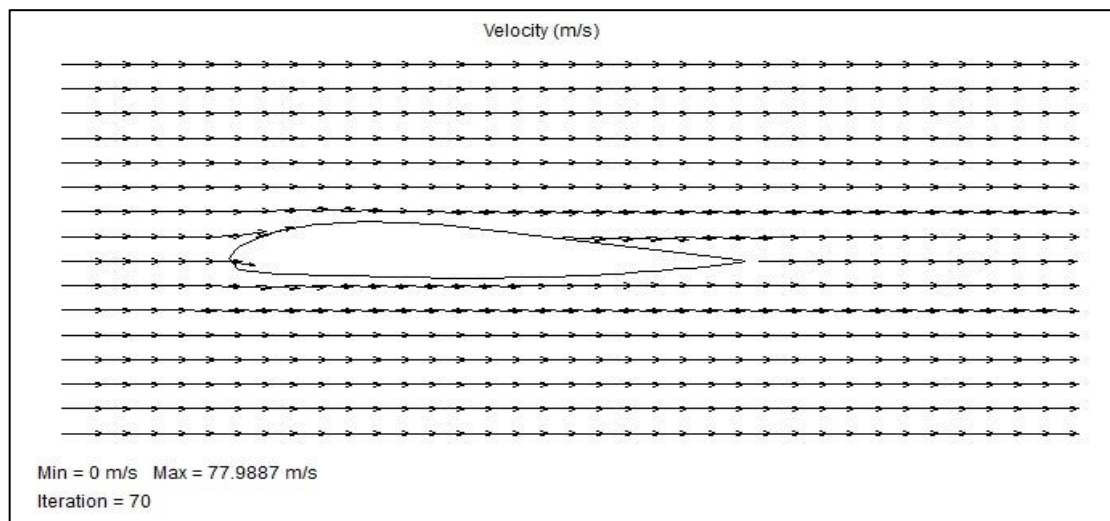
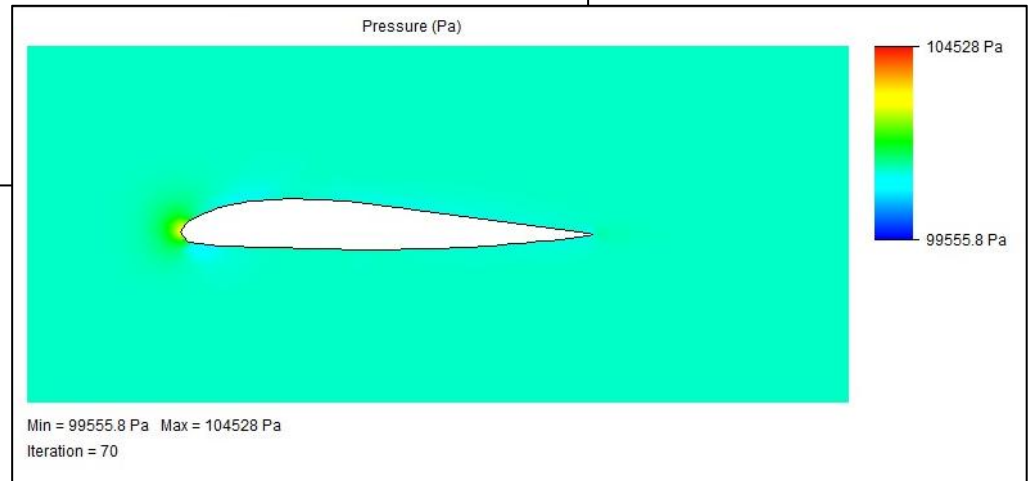
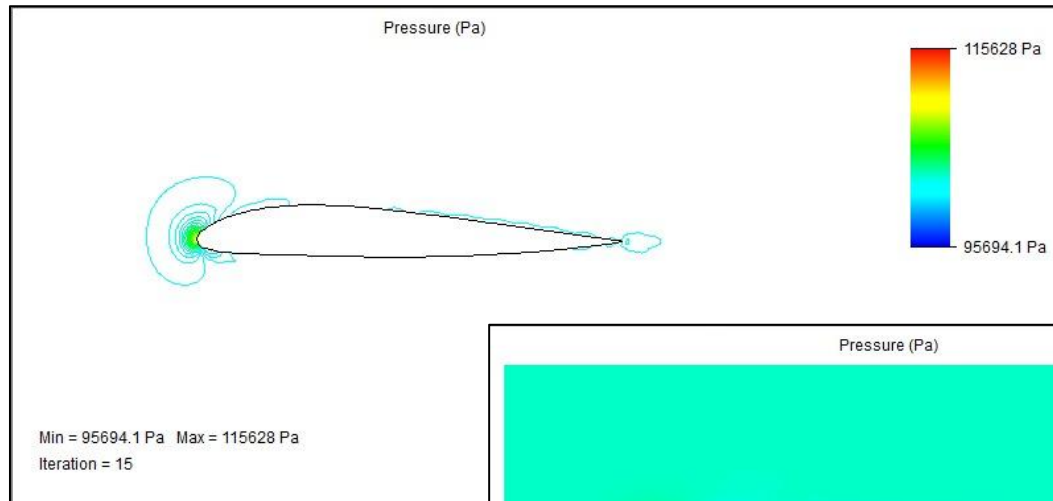
200K Re



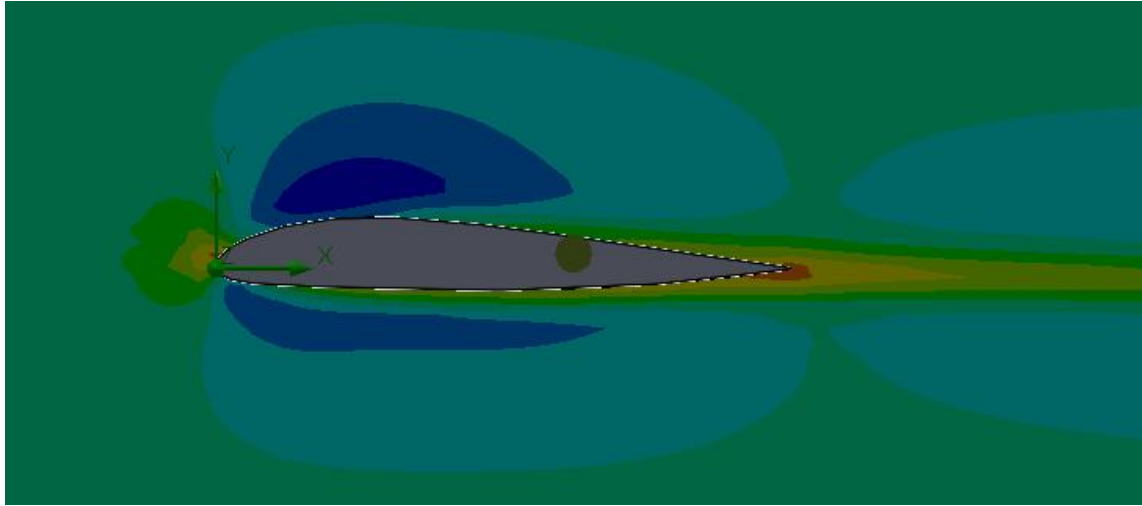
1M Re



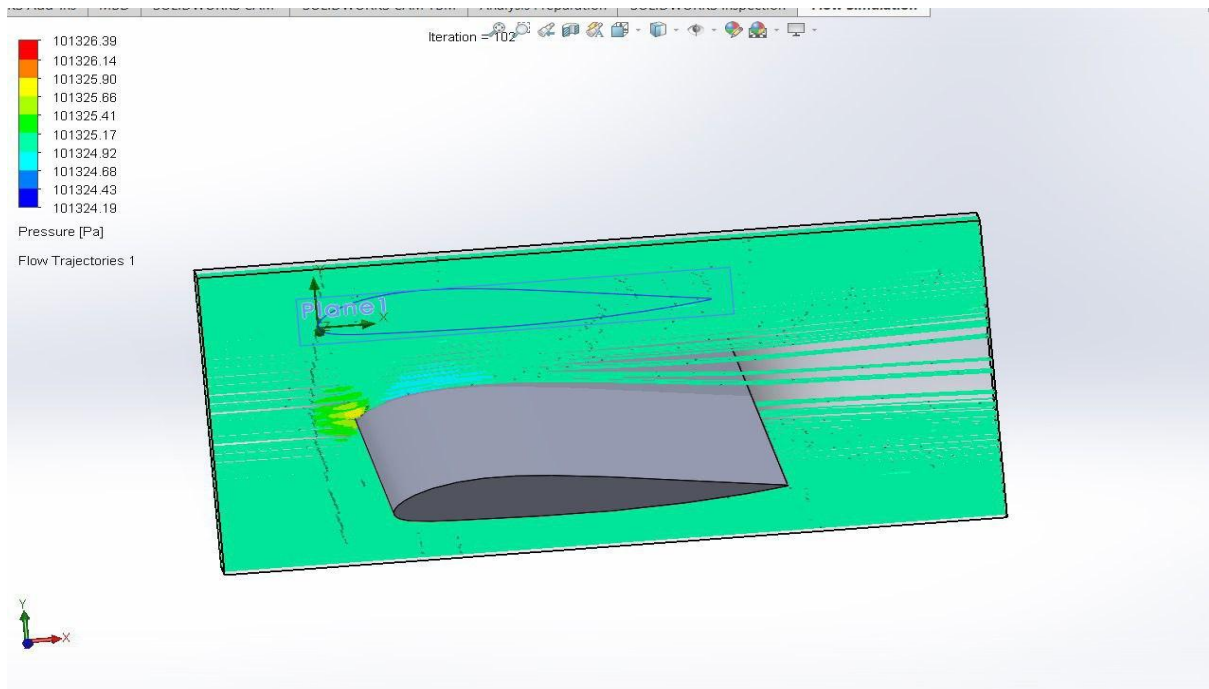
2M Re



Temperature:



Flow trajectory:



We used the computational domain to get less processing time and less load on the PC but in real life projects it's not considered a real flow simulation.

For more info and graphs you can check the excel sheets

GOAL PLOT FROM EXCLE:

20K

Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
SG Average Static Pressure 1	[Pa]	101324.976	101324.9758	101324.9756	101324.9761	100	Yes	0.000522	0.001013
Drag Force (X) 2	[N]	0.000550976	0.000550216	0.000548888	0.000552344	100	Yes	1.26E-06	8.57E-06
Left Force (Y) 3	[N]	-5.74839E-05	-6.07698E-05	-6.58705E-05	-5.7094E-05	100	Yes	3.9E-06	3.99E-06
Drag Coefficient	[]	0.046961646	0.046896811	0.046783693	0.047078237	100	Yes	0.000108	0.000731
Left Coefficient	[]	-0.019702598	-0.020828863	-0.022577119	-0.01956886	100	Yes	0.001335	0.001366

50K

Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
SG Average Static Pressure 1	[Pa]	101323.8719	101323.8654	101323.8435	101323.8779	29.3	Yes	0.011156	0.003271
Drag Force (X) 2	[N]	0.00057297	0.000579402	0.000560138	0.000620457	100	Yes	7.94E-06	9.25E-06
Left Force (Y) 3	[N]	0.000893157	0.000842617	0.000485958	0.001064905	18	Yes	0.000127	2.3E-05
Drag Coefficient	[]	0.028418057	0.02873706	0.027781609	0.030773275	100	Yes	0.000394	0.000459
Left Coefficient	[N]	0.044298621	0.041791941	0.024102441	0.05281691	18	Yes	0.0063	0.001139

100K

#NAME?

Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
Drag Force	[N]	0.017157237	0.017504813	0.017157237	0.017794313	100	Yes	0.000383	0.000386
Left Force	[N]	0.017053621	0.01734884	0.017053621	0.01751946	100	Yes	0.000305	0.000407
SG Average Static Pressure 3	[Pa]	101323.3656	101323.3191	101323.2864	101323.3656	100	Yes	0.015571	0.017457
Drag Coefficient	[]	0.027464316	0.028020696	0.027464316	0.02848411	100	Yes	0.000614	0.000618
Left Coefficient	[]	0.027298454	0.027771023	0.027298454	0.028044143	100	Yes	0.000489	0.000652

200K

Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
SG Average Static Pressure 1	[Pa]	101322.7642	101323.0515	101321.8944	101335.9302	0	Yes	12.87871	0
Drag Force (X) 2	[N]	0.022074302	0.066470679	0.021774675	1.215598046	0	Yes	1.149127	0
Lift Force (Y) 3	[N]	-0.010078699	-0.001019402	-0.041725963	0.250335733	0	Yes	0.251355	0
Drag coefficient	[]	0.027751479	0.083565934	0.027374793	1.528231493	0	Yes	1.444666	0
Lift coefficient	[N]	-0.012670788	-0.001281577	-0.05245725	0.3147183	0	Yes	0.316	0

1M

Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
SG Average Static Pressure 1	[Pa]	101289.082	101288.9225	101287.8911	101289.3992	100	Yes	1.508027	1.712381
drag Force (X) 2	[N]	0.752872837	0.753363966	0.75222556	0.755713281	100	Yes	0.003488	0.005597
lift Force (Y) 3	[N]	-0.143423575	-0.143072222	-0.146115109	-0.14166348	100	Yes	0.004452	0.009254
drag coeff	[]	0.0011358	0.001136541	0.001134824	0.001140086	100	Yes	5.26E-06	8.44E-06
lift coeff	[N]	-0.000216372	-0.000215842	-0.000220432	#NAME?	100	Yes	6.72E-06	1.4E-05

2M

Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
SG Average Static Pressure 1	[Pa]	101250.9974	101247.5694	101235.2934	101250.9974	100	Yes	15.70397	44.09743
drag force	[N]	2.374387606	2.374603125	2.373507653	2.380215992	100	Yes	0.006708	0.358341
lift force	[N]	-0.595041721	-0.593192047	-0.597021426	-0.581459	100	Yes	0.015562	0.298931
drag coefficient	[N]	0.03796945	0.037972897	0.037955379	0.038062653	100	Yes	0.000107	0.00573
lift coefficient	[N]	-0.009515467	-0.009485888	-0.009547125	-0.00929826	100	Yes	0.000249	0.00478

Discussing the graphs:

Figure (1)

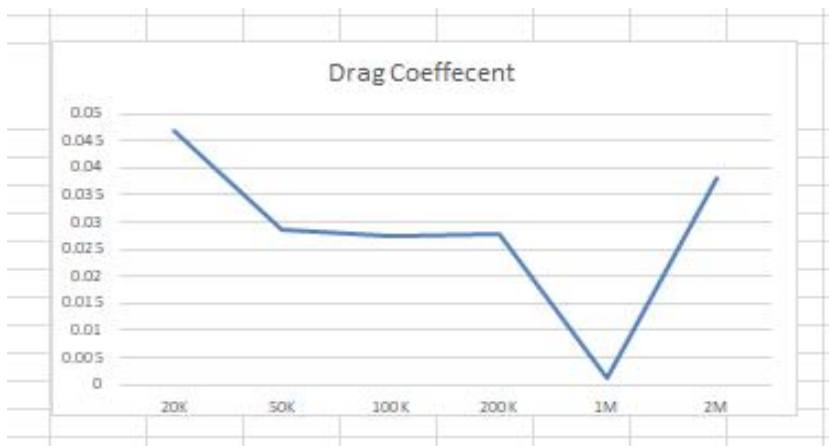


Figure (2)

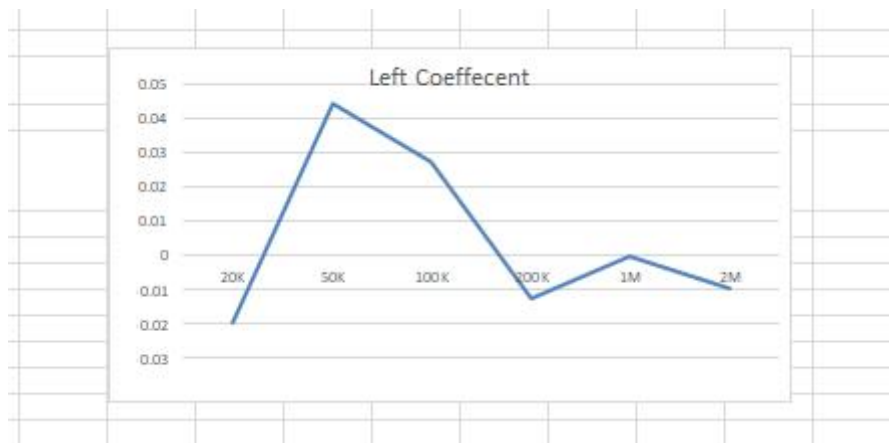


Figure (3)

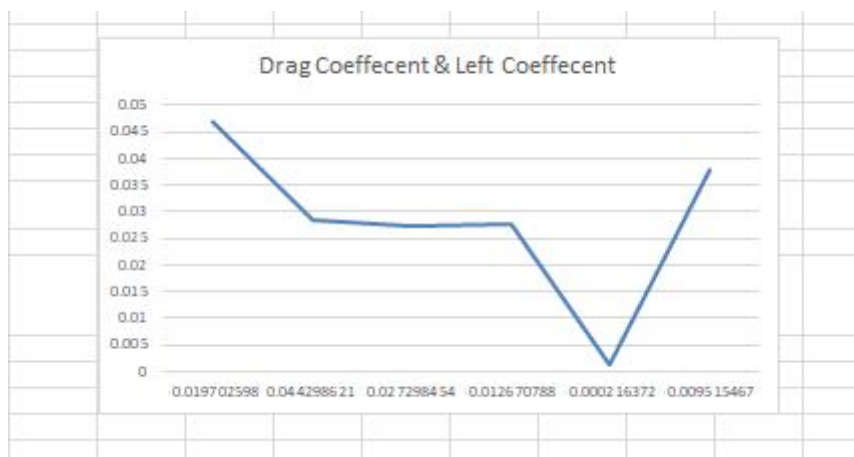


Figure (4)

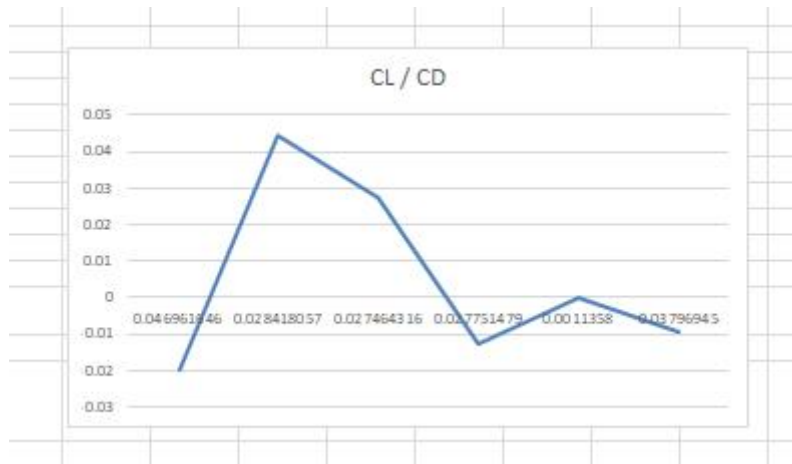
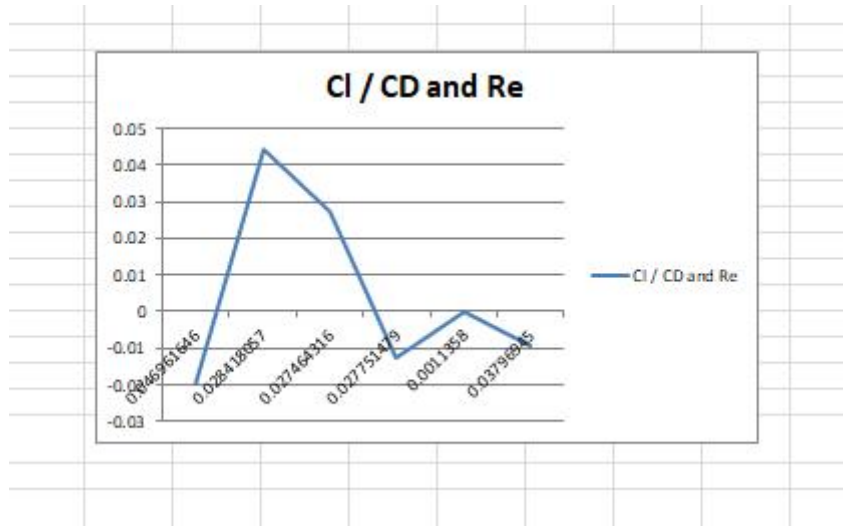


Figure (5)



20K.xlsx



50K.xlsx



100K.xlsx



200K.xlsx



1M.xlsx



2M.xlsx



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“if you face any problem in opening files here is the [Drive link](#)”

Table of the resulting data

AoA	Re	velocity	FL	CL	FD	CD
0	20K	0.1478 m/s	-5.7484E-05	-0.0197026	0.000550976	0.046961646
	50K	0.3694 m/s	0.000893157	0.044298621	0.00057297	0.028418057
	100K	0.7388 m/s	0.017053621	0.027298454	0.017157237	0.027464316
	200K	1.4776 m/s	0.028762838	0.036160206	0.013077891	0.016441326
	1M	7.3878 m/s	0.767784935	0.001158297	0.702906816	0.001060421
	2M	14.7755 m/s	-0.59504172	-0.00951547	2.374387606	0.03796945

Conclusion

- Simulation focuses on steady-state condition with fixed AoA = 0°
- Lift expected to be minimal due to zero AoA
- Drag and surface pressure distribution observed
- Velocity varies with Re, affecting flow separation and pressure